



Homework 8: Breit-Wigner resonance and Gadolinium.

This question will explore transitions from an initial state $|k_i\rangle$ to a final state $|k_f\rangle$ that proceed through an intermediate resonance $|R\rangle$, which is an unstable state with complex energy $\mathcal{E}_R$.

This is a continuation of HW 7.

Consider the non-relativistic scattering of two particles through an intermediate state $|R\rangle$ of H_0 (R for “resonance”). Working in the center-of-mass frame, we can treat the problem as a single particle of reduced mass μ scattering off a potential V that allows for the resonance $|R\rangle$ to be created or destroyed. The scattering states are plane waves in a periodic box of volume L^3 ,

$$\langle \mathbf{r} | \mathbf{k} \rangle = \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{L^{3/2}}, \quad (1)$$

so that $\langle \mathbf{k}' | \mathbf{k} \rangle = \delta_{\mathbf{k}'\mathbf{k}}$ and the density of states in k -space is $L^3/(2\pi)^3$. The resonance $|R\rangle$ is a discrete, unit-normalized eigenstate of H_0 . Assume for simplicity that $\langle R | V | \mathbf{k} \rangle$ depends only on $|\mathbf{k}|$ and not on $\hat{\mathbf{k}}$ (the “s-wave” nomenclature comes from the partial wave expansion of the scattering amplitude, where $\ell = 0$ corresponds to the spherically symmetric partial wave (s-wave), $\ell = 1$ to the p-wave, $\ell = 2$ to the d-wave, and so on).

From the second-order Dyson series, the transition amplitude from $|\mathbf{k}_i\rangle$ to $|\mathbf{k}_f\rangle$ through the resonance is

$$T_{fi}^{(\text{box})} = \frac{\langle \mathbf{k}_f | V | R \rangle \langle R | V | \mathbf{k}_i \rangle}{E - \mathcal{E}_R}, \quad (2)$$

where $\mathcal{E}_R = E_R + \Delta E_R - i\Gamma/2$ is the complex energy from HW 7 part (d). We define the normalized matrix element

$$\bar{V}_{R\mathbf{k}} \equiv L^{3/2} \langle R | V | \mathbf{k} \rangle. \quad (3)$$

This is just the matrix element taken with the plane wave $e^{i\mathbf{k}\cdot\mathbf{r}}$ rather than $e^{i\mathbf{k}\cdot\mathbf{r}}/L^{3/2}$.

a) Show that the box-normalized matrix element squared is

$$|T_{fi}^{(\text{box})}|^2 = \frac{|\bar{V}_{R\mathbf{k}_f}|^2 |\bar{V}_{R\mathbf{k}_i}|^2}{L^6 [(E - \tilde{E}_R)^2 + \Gamma^2/4]}, \quad (4)$$

where $\tilde{E}_R \equiv E_R + \Delta E_R$.

b) A resonance $|R\rangle$ may decay into several distinct sets of final states, called “decay channels.” For example, the excited nucleus $^{17}\text{O}^*$ can be produced and can decay through channels such as $n + ^{16}\text{O}$ (channel a) or $p + ^{16}\text{N}$ (channel b). The process $n + ^{16}\text{O} \rightarrow ^{17}\text{O}^* \rightarrow p + ^{16}\text{N}$ is then a scattering from channel a to channel b through the resonance. Each channel has its own coupling to $|R\rangle$ and its own “partial width” $\Gamma_a, \Gamma_b, \dots$, defined as the decay rate of $|R\rangle$ into that channel alone (for example, Γ_a is the decay rate of $^{17}\text{O}^*$ into $n + ^{16}\text{O}$ and Γ_b is the decay rate into $p + ^{16}\text{N}$). The total width is

$$\Gamma = \Gamma_a + \Gamma_b + \dots \quad (5)$$

We may also define a branching ratio

$$\text{BR}_a = \frac{\Gamma_a}{\Gamma}, \quad (6)$$

where it should be clear that $\sum_a \text{BR}_a = 1$. Clearly, our example b is inelastic and ought to have a different partial width Γ_b than the elastic example a . In what follows, however, we will assume

that all channels are elastic, so that the initial and final momenta are the same, $k_i = k_f = k$.¹

The decay rate of $|R\rangle$ into all states of a particular channel is still given by Fermi's Golden Rule,

$$\Gamma_a = 2\pi \sum_{\mathbf{k}} |\langle \mathbf{k} | V | R \rangle|^2 \delta(E_R - E_k) \longrightarrow \frac{\mu k_R}{\pi} |\bar{V}_R|_a^2, \quad (7)$$

where $E_k = k^2/(2\mu)$ and $k_R = \sqrt{2\mu E_R}$. We keep the subscript a to indicate that the matrix element is for the channel a . For another channel b , Γ_b takes the same form with the appropriate matrix element $|\bar{V}_R|_b^2$. Show the correspondence in Eq. (7) by replacing the sum over $\mathbf{k}$ with an integral and using the density of states in k -space. Assume that the matrix element is independent of the direction of $\mathbf{k}$, so that the integral over d^3k reduces to an integral over dk , which you can evaluate using the delta function. Then express $|\bar{V}_{Rk_i}|^2$ and $|\bar{V}_{Rk_f}|^2$ in terms of the partial widths Γ_a and Γ_b .

(Hint: use the property of the delta function $\delta(f(k)) = \sum_{k_0} \delta(k - k_0)/|f'(k_0)|$ with k_0 the roots of f .)

c) Apply Fermi's Golden Rule for the scattering rate into a solid angle $d\Omega_f$:

$$dw = 2\pi |T_{fi}^{(\text{box})}|^2 \rho_f(E) d\Omega_f, \quad \rho_f(E) = \frac{L^3 \mu k_f}{(2\pi)^3}, \quad (8)$$

where $\rho_f(E)$ is the density of final states per unit energy per unit solid angle. Dividing by the incident flux $\Phi = v_i/L^3 = k_i/(\mu L^3)$, derive the **differential** cross section, verifying that all factors of L cancel:

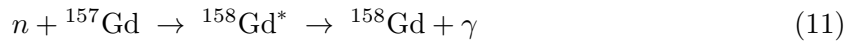
$$\frac{d\sigma}{d\Omega_f} = \frac{\mu^2}{4\pi^2} \frac{|\bar{V}_{Rk_f}|^2 |\bar{V}_{Rk_i}|^2}{(E - \tilde{E}_R)^2 + \Gamma^2/4}. \quad (9)$$

Then integrate over $d\Omega_f$ (using the s-wave isotropy and $k_f = k_i = k$ for elastic scattering) and express the result in terms of the partial widths from part (b). Show that the total cross section is

$$\sigma(E) = \frac{\pi}{k^2} \frac{\Gamma_a \Gamma_b}{(E - \tilde{E}_R)^2 + \Gamma^2/4} \quad (10)$$

where $\Gamma = \Gamma_a + \Gamma_b + \dots$ is the total width.

(d) Gadolinium-157 has the largest thermal neutron capture cross section of any stable nuclide, a property exploited in nuclear reactor control rods and in gadolinium-doped water Cherenkov neutrino detectors (such as the Super-Kamiokande experiment). The process



proceeds through a compound nuclear resonance at $E_R = 0.0314$ eV. The entrance channel (a) is the neutron and the exit channel (b) is the photon. The resonance parameters are²

$$\Gamma_n^{\text{eff}} = g\Gamma_n^0 \sqrt{E_R/\text{eV}} = 0.590 \text{ meV} \times \sqrt{0.0314} \approx 0.105 \text{ meV}, \quad \Gamma_\gamma = 106 \text{ meV}, \quad \Gamma \approx \Gamma_\gamma. \quad (12)$$

¹Technically, for any channel the resonance energy E_R is the same, but the momentum k is different because of the different reduced mass μ in each channel. We ignore this subtlety and assume that k is the same for all channels, though it can be easily generalized by noting that if a given channel has a threshold energy $Q_{a,b}$, then $k_{a,b} = \sqrt{2\mu_{a,b}(E_R - Q_{a,b})}$. For the process to be kinematically allowed, we need $E_R > Q_{a,b}$, so that $k_{a,b}$ is real.

²The effective neutron width $\Gamma_n^{\text{eff}} \equiv g_J \Gamma_n$ absorbs the spin statistical factor $g_J = (2J+1)/[(2s+1)(2I+1)] = 5/8$, which counts the fraction of incident spin channels that can form the $J^\pi = 2^-$ compound state from an s-wave neutron ($s = 1/2$) on a ${}^{157}\text{Gd}$ target ($I = 3/2$). This detail is needed for unpolarized beams, but the Breit-Wigner formula from Eq. (10) applies as written with $\Gamma_a = \Gamma_n^{\text{eff}}$. The tabulated value $g\Gamma_n^0 = 0.590$ meV is the *reduced* width at a reference energy of 1 eV; the physical width at the resonance energy is $\Gamma_n^{\text{eff}}(E_R) = g\Gamma_n^0 \sqrt{E_R/1 \text{ eV}}$.

In the Breit–Wigner formula (10), use $\Gamma_a = \Gamma_n^{\text{eff}}$ and $\Gamma_b = \Gamma_\gamma$. The reduced mass is $\mu \approx m_n(1 - 1/(A + 1)) \approx 933.6 \text{ MeV}$, with $A = 157$.

Plot $\sigma(E)$ in the range $E = 0.001\text{--}1 \text{ eV}$. Overlay the unitarity bound $4\pi/k^2$ and the geometric cross section πR_{nuc}^2 with $R_{\text{nuc}} = 1.2 A^{1/3} \text{ fm}$. Use logarithmic axes.

(Bonus question) In a water Cherenkov detector, an inverse beta decay event ($\bar{\nu}_e + p \rightarrow e^+ + n$) produces a neutron that thermalizes and is subsequently captured. Capture on hydrogen ($n + p \rightarrow d + \gamma$, $E_\gamma = 2.2 \text{ MeV}$) produces a single low-energy photon that is difficult to detect. Capture on gadolinium produces a γ -ray cascade totaling $\sim 8 \text{ MeV}$, which is much easier to identify. This motivates doping the water with a small amount of gadolinium.

Consider water doped with 0.01% Gd by mass (the approximate loading of Super-Kamiokande). The number density of ^{157}Gd atoms in this solution is $n_{157} \approx 6.0 \times 10^{16} \text{ cm}^{-3}$.³ The number density of hydrogen atoms in water is $n_H \approx 6.7 \times 10^{22} \text{ cm}^{-3}$, and the thermal neutron capture cross section on hydrogen is $\sigma_H \approx 0.33 \text{ barn}$.

- (i) Compute the macroscopic cross sections $\Sigma_{\text{Gd}} = n_{157} \sigma_{\text{Gd}}$ and $\Sigma_H = n_H \sigma_H$ at thermal energies. Use the single-resonance σ_{Gd} from part (d).
- (ii) The mean free path for neutron capture is $\lambda = 1/(\Sigma_H + \Sigma_{\text{Gd}})$. Compute λ with and without Gd doping.
- (iii) What fraction of neutrons are captured by ^{157}Gd rather than by hydrogen? Why is this advantageous for detecting reactor antineutrinos?

³This follows from $n_{157} = f_{\text{Gd}} \eta_{157} \rho_w N_A / A_{\text{Gd}}$, where $f_{\text{Gd}} = 10^{-4}$ is the Gd mass fraction, $\eta_{157} = 15.65\%$ is the isotopic abundance, $\rho_w \approx 1 \text{ g/cm}^3$, N_A is Avogadro's number, and $A_{\text{Gd}} \approx 157 \text{ g/mol}$.